

东昆仑西段夏日哈木榴辉岩原岩属性及陆(弧)陆碰撞

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内容提要:东昆仑西段夏日哈木榴辉岩呈透镜状分布于古元古代金水口群中,榴辉岩的 SiO_2 含量 47.4%~50.78%, $\text{Mg}^\#$ 值为 40~62,表明其原岩为分异程度较高的基性岩,属于拉斑玄武岩系列。基于岩石 Nb 的含量,将夏日哈木榴辉岩分为两类——富 Nb 型榴辉岩($\text{Nb}>7\times 10^{-6}$)和正常榴辉岩($\text{Nb}<7\times 10^{-6}$)。富 Nb 型榴辉岩具有贫 SiO_2 、 MgO 、 Al_2O_3 ,富集 $\text{FeO}^\dagger$ 、 TiO_2 的特点;稀土元素配分曲线具有右倾的 OIB 特点,与古生代富 Nb 玄武岩和 Seki 玄武岩微量元素特征类似;微量元素具有弱的 Nb、Ta、Ti 的负异常,与古生代富 Nb 玄武岩特征类似。正常榴辉岩具有贫 $\text{FeO}^\dagger$ 、 TiO_2 ,富集 SiO_2 、 MgO 、 Al_2O_3 的特点;稀土元素配分曲线具有弱右倾的 E-MORB 特点,微量元素具有弱的 Nb、Ta、P 的负异常,U、Sr 的正异常,类似于 Okinawa 板内弧后盆地玄武岩微量元素特征。富 Nb 型榴辉岩的 $\epsilon_{\text{Nd}}(0)$ 和 $\epsilon_{\text{Nd}}(t=571\text{Ma})$ 值分别为 0~4.55、-0.88~7.26,正常榴辉岩的 $\epsilon_{\text{Nd}}(0)$ 和 $\epsilon_{\text{Nd}}(t=571\text{Ma})$ 值分别为 2.2~5.19、3.17~6.22,两者均位于加里东 HP-UHP 变质带和大别-苏鲁 HP-UHP 变质带榴辉岩 Sm-Nd 同位素组成范围内。在微量元素图解上,样品显示具有地壳混染的趋势。由此可知,夏日哈木榴辉岩的原岩可能是具有陆内弧后盆地玄武岩性质的岩石,源区受到俯冲板片熔体的改造,岩浆过程中同化混染一定量的下地壳。根据现有的年代学资料,夏日哈木榴辉岩峰期变质可能发生在晚奥陶世—志留纪。夏日哈木榴辉岩可能是在弧后盆地消亡过程中,弧陆发生俯冲碰撞的结果。

关键词:陆(弧)-陆碰撞;榴辉岩;富 Nb 玄武岩;夏日哈木;东昆仑

榴辉岩是一种出现在造山带中的高压/超高压变质岩(Smith, 1988; Carswell, 1990),通常能够指示在汇聚板块边缘存在洋壳或陆壳的深俯冲作用(Dobretsov, 1991; Ernst and Liou, 1999; Chopin, 2003),对研究大陆增生和造山过程有着十分重要的意义(Zheng Yongfei, 2012; Klemd, 2013)。Maruyama et al. (1996)将高压—超高压变质带划分为阿尔卑斯型和太平洋型,前者以大陆地壳的俯冲折返为特征,代表陆陆碰撞造山作用,后者以大洋地壳俯冲折返为特征,代表增生造山作用(Cawood et al., 2003)。这两种高压—超高压变质带的差异主要体现在榴辉岩等变质岩的原岩性质、变质条件以及共生组合上(Zhang Lifei et al., 2008; Liou et al., 2009; Zheng Yongfei et al., 2012)。若榴辉岩的原岩具有洋壳性质,代表了洋壳的俯冲变质,如中国的西南天山(Gao Jun and Klemd, 2003)、北祁连和北阿尔金(Zhang Jianxin et al., 2006)以及阿尔卑斯造山带和海西造山带榴辉岩(Bernard-Griffiths

and Cornichet, 1985; Jahn, 1999);若其原岩具有陆壳性质,代表了大陆地壳的俯冲作用,如中国的苏鲁-大别造山带(Wang Shiguang et al., 1995; Chavagnac and Jahn, 1996; Liou and Zhang, 1998; Jahn, 1998)、柴北缘北带(Song Shuguang et al., 2003; Zhang Jianxin et al., 2006; Mattinson et al., 2007)和加里东造山带榴辉岩(Griffin and Brueckner, 1980; Jamtveit et al., 1991)。因此,榴辉岩的原岩恢复对认识造山带性质和造山过程具有十分重要的意义(Wang Hao et al., 2013; Meng Fancong et al., 2013, 2015)。

东昆仑蛇绿岩(Yang Jingsui et al., 1996; Shen Yuanchao et al., 2000; Liu Zhanqing et al., 2011)、花岗岩(Mo Xuanxue et al., 2007; Li Ruibao et al., 2013a, 2013b, 2017; Li Zuochen et al., 2014; Hao Nana et al., 2014; Chen Guochao et al., 2017b; Jian Kunkun et al., 2017)、麻粒岩(Zhang Jianxin et al., 2003; Li Huaikun et al.,

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2006)以及榴辉岩(Meng Fancong et al., 2013, 2015; Qi Shengsheng et al., 2014; Qi Xiaopeng et al., 2016; Du Wei et al., 2017)的研究表明东昆仑存在早古生代造山作用(Jiang Chunfa et al., 1992; Pan Yusheng et al., 1996; Yin Fuguang et al., 2003),东昆中断裂带也被认为是一条早古生代缝合带(Xiao Xuchang et al., 1986; Gao Yanlin et al., 1988; Chen Bingwei et al., 1996; Lu Songnian et al., 2006)。然而,对于东昆仑的造山过程还存在不同的认识,大部分学者倾向认为是增生造山(Yin An, 2001; Bian Qiantao et al., 2002; Pan Guitang et al., 2012),但缺少洋壳俯冲变质的岩石学证据;亦有部分学者认为是陆陆碰撞造山(Meng Fancong et al., 2013, 2015; Qi Shengsheng et al., 2014)。因此,对东昆仑早古生代造山过程仍需要深入的研究(Li Rongshe et al., 2007)。东昆仑东段温泉榴辉岩代表了俯冲的陆壳,证明了东昆仑早古生代发生过陆陆碰撞造山作用(Jia Lihui et al., 2014; Meng Fancong et al., 2015),但是东昆仑西段榴辉岩代表了俯冲的洋壳还是陆壳仍不清楚。本文依据东昆仑西段夏日哈木榴辉岩的地球化学和 Sm-Nd 同位素对榴辉岩原岩进行恢复,进一步讨论东昆仑早古生代造山过程和性质。

1 地质背景

东昆仑造山带位于青藏高原北部,柴达木陆块和松潘-甘孜陆块之间,呈近东西向展布,其西段以阿尔金断裂为界,东部大体以温泉-赛什塘断裂与秦岭造山带相连(图 1a)(Jiang Chunfa et al., 1992; Pan Guitang et al., 2002; Zhang Xueting et al., 2007)。东昆仑以昆中断裂带为界可分为昆北地体和昆南地体两部分(Wu Gongjian et al., 1989; Xu Zhiqin et al., 2006)。

昆北地体以广泛分布太古宙(?)—元古宙变质基底为特征,出露有太古宙(?)—古元古代金水口群和中元古代的小庙岩组(Wang Yunshan et al., 1987; Bureau of Qinghai Geology and Mineral Resources, 1991; Lu Songnian et al., 2002; Wang Guocan et al., 2004, 2007; He Dengfeng et al., 2016),新元古代—早古生代浅变质的万宝沟群和纳赤台群(Jiang Chunfa et al., 1992),志留纪—泥盆纪花岗闪长岩、花岗岩和泥盆纪的陆相火山岩(Li Ruibao et al., 2013a; Li Zuochun et al., 2014; Hao Nana et al., 2014; Xu Zhiqin et al., 2007; Lu

Lu et al., 2010; Cui Meihui et al., 2011),以及大面积的印支期花岗闪长岩、花岗岩(Mo Xuanxue et al., 2007; Xu Zhiqin et al., 2007; Ren Haidong et al., 2016)和晚三叠世鄂拉山组大陆边缘火山岩(Liu Hongtao, 2001)。沿昆中断裂带出露的超基性岩、辉长岩、辉绿岩和基性火山岩等岩片或岩块,混杂在前寒武纪变质岩系和早古生代纳赤台群中(Jiang Chunfa et al., 1992)。大部分学者认为这些岩块是被肢解的蛇绿岩套,主要形成于早古生代,是原特提斯洋的残片(Xiao Xuchang et al., 1986; Gao Yanlin et al., 1988; Pan Yusheng et al., 1996; Chen Bingwei et al., 1996; Yang Jingsui et al., 1996; Zhu Yunhai et al., 1999, 2006; Cui Meihui et al., 2011)。近年来,在昆北地体的变质基底中相继发现了温泉榴辉岩、夏日哈木榴辉岩、朗木日榴辉岩、大格勒榴辉岩等,形成时代分别为 $428 \pm 2\text{Ma}$ (Meng Fancong et al., 2013)、 $411 \pm 1.9\text{Ma}$ (Qi Shengsheng et al., 2014)、 $431.9 \pm 2.3\text{Ma}$ (Qi Xiaopeng et al., 2016)和 $433 \pm 5\text{Ma}$ (Du wei et al., 2017),是晚奥陶世—早泥盆世陆陆碰撞的产物(Meng Fancong et al., 2013, 2015)。

昆南地体由前寒武纪的苦海群和万宝沟群、早古生代纳赤台群、晚泥盆世红色磨拉石建造,石炭纪—中三叠世浅海相地层以及晚三叠世—侏罗纪陆相地层组成(Jiang Chunfa et al., 1992; Pan Yusheng et al., 1996; Wang Guocan et al., 2004, 2007; Liu Yongjiang et al., 2005)。其中苦海群由角闪岩相变质的斜长角闪岩-长英质片麻岩及大理岩构成,原岩为碎屑岩-中基性火山岩-碳酸盐岩(Liu Yongjiang et al., 2005);万宝沟群原岩为一套碎屑岩、大理岩和火山岩组合,遭受绿片岩相变质(Bureau of Qinghai Geology and Mineral Resources, 1991; Jiang Chunfa et al., 1992; Pan Yusheng et al., 1996)。上述地层部分被晚三叠世花岗岩侵入(Mo Xuanxue et al., 2007)。沿昆南断裂由西向东分布的蛇绿岩有大九坝、黑茨沟、布青山、下大武、玛积雪山和玛沁蛇绿岩(Jiang Chunfa et al., 1992; Yang Jingsui et al., 1996, 2004, 2009; Chen Liang et al., 2000; Bian Qiantao et al., 2004; Guo Anlin et al., 2006),这些蛇绿岩主要形成于晚古生代,代表了古特提斯洋的残片(Chen Liang et al., 2000; Konstantinovskaia et al., 2003; Bian Qiantao et al., 2004; Yang Jingsui et al., 2009; Liu Zhanqing et al., 2011; Xiong Fuhao et al., 2014;

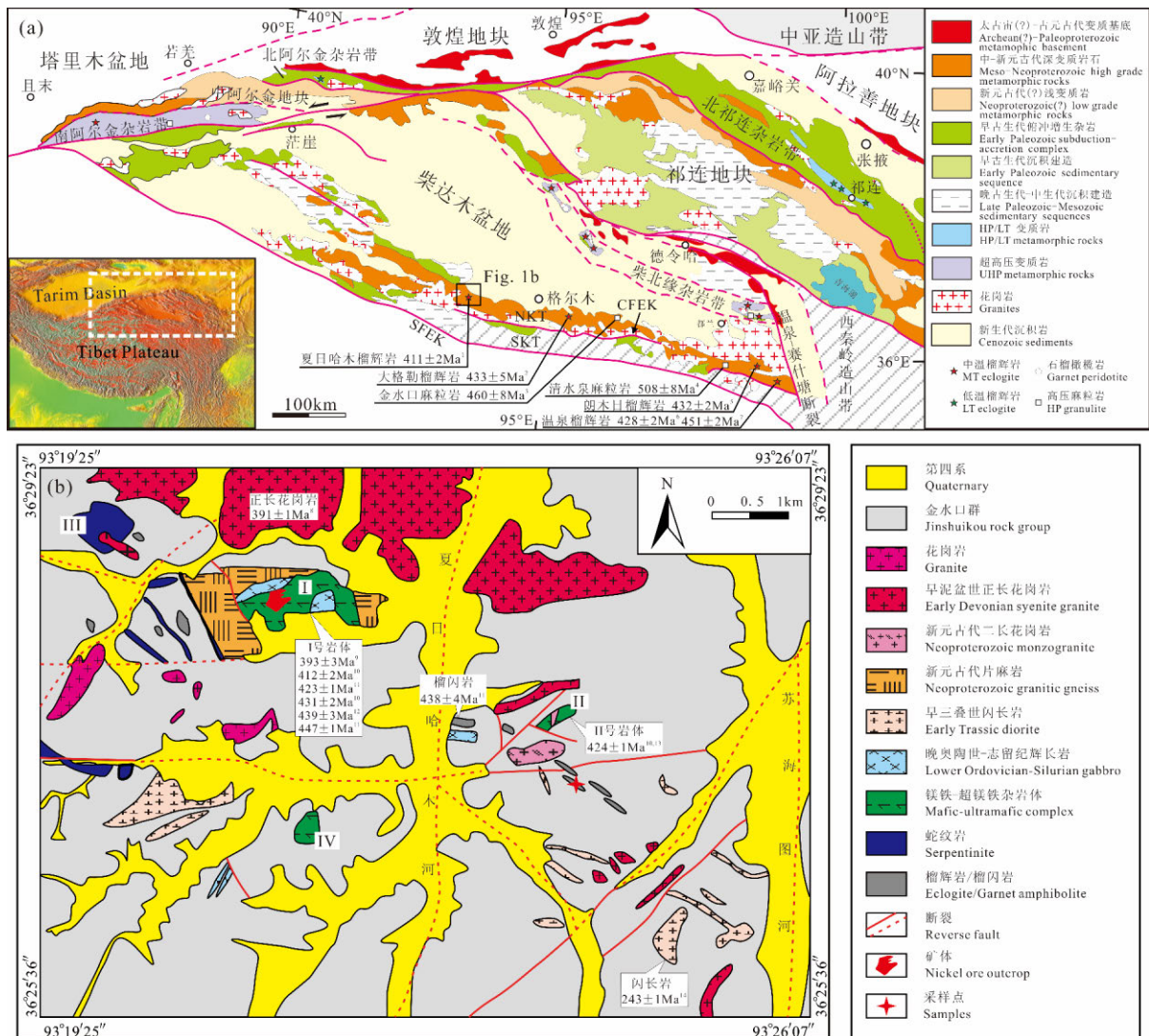


图 1 青藏高原北缘构造略图(a)(据 Zhang Jianxin et al., 2015)及夏日哈木地区地质简图(b)
(据 Li Shijin et al., 2012; Wang Guan, 2014)

Fig. 1 Structural sketch of the northern Tibet (a) (modified from Zhang Jianxin et al., 2015) and geological sketch map of the Xiarihamu area (b) (modified from Li Shijin et al., 2012 and Wang Guan, 2014)

CFEK—东昆中断裂, SFEK—东昆南断裂, NKT—昆北地体, SKT—昆南地体; 年代学资料来自: 1—祁生胜等, 2014; 2—Du Wei et al., 2017; 3—张建新等, 2003; 4—李怀坤等, 2006; 5—祁晓鹏等, 2016; 6—Meng Fancong et al., 2013; 7—贾丽辉等, 2014; 8—王冠等, 2013; 9—李世金等, 2012; 10—Li CS et al., 2015; 11—王冠, 2014; 12—姜常义等, 2015; 13—Peng Bo et al., 2016; 14—王冠等, 2014

CFEK—Central Fault of East Kunlun, SFEK—South Fault of East Kunlun, NKT—North Kunlun terrane, SKT—South Kunlun terrane; Age data source: 1—Qi Shengsheng et al., 2014; 2—Du Wei et al., 2017; 3—Zhang Jianxin et al., 2003; 4—Li Huaikun et al., 2006; 5—Qi Xiaopeng et al., 2016; 6—Meng Fancong et al., 2013; 7—Jia Lihui et al., 2014; 8—Wang Guan et al., 2013; 9—Li Shijin et al., 2012; 10—Li CS et al., 2015; 11—Wang Guan, 2014; 12—Jiang Changyi et al., 2015; 13—Peng Bo et al., 2016; 14—Wang Guan et al., 2014

Chen Guochao et al., 2017a)。

2 榴辉(闪)岩产状及岩石学

东昆仑夏日哈木榴辉岩、榴闪岩分布在夏日木河两侧, 断续出露长约 8km, 均呈透镜体和不规则状分布于金水口岩群条带状黑云斜长片麻岩夹角闪片岩中(图 1b)。在夏日哈木河附近, 榴辉(闪)岩呈

大小不等的透镜体出露, 近北西西或东西向分布, 最大透镜体长约 30m、宽约 12m, 最小者长约 2m、宽约 1.3m(Qi Shengsheng et al., 2014)。在夏日哈木矿区以东的苏海图、以西的拉陵灶火一带, 也分布着榴闪岩、榴辉岩, 它们延续长达 20km(Qi Shengsheng et al., 2014)。夏日哈木榴辉岩大部分已经退变为榴闪岩, 灰黑色—灰绿色, 中—细粒变斑

晶结构、“白眼圈”结构,块状构造(图 2a)。变斑晶主要为石榴子石,含量约为 10%~35%,粒度变化较大(0.1~6mm)(图 2b,c),成分以铁铝榴石端元为主(Du Wei, 2015)。基质含量 60%~80%,主要由斜长石和角闪石构成的后成合晶(图 2d)以及少量残余的单斜辉石、透辉石和不透明矿物组成(Qi Shengsheng et al., 2014)。Qi Shengsheng et al (2014)通过矿物学研究确定了榴辉岩相的变质矿物组合为石榴子石+绿辉石+金红石+钛铁矿,榴闪

岩的矿物组合为普通角闪石+斜长石+透辉石,并估算出榴辉岩形成时的下限温压条件为 $T=660\sim 700^{\circ}\text{C}$ 、 $P=20\text{kbar}$,榴闪岩的温压条件为 $T=550^{\circ}\text{C}$ 、 $P=7\text{kbar}$ 。

3 测试方法

岩石全分析在国家地质实验测试中心完成,主量元素采用 X-荧光光谱法,测试仪器为 3080E 型 X-荧光光谱仪,其中 FeO 采用化学滴定法测得;稀

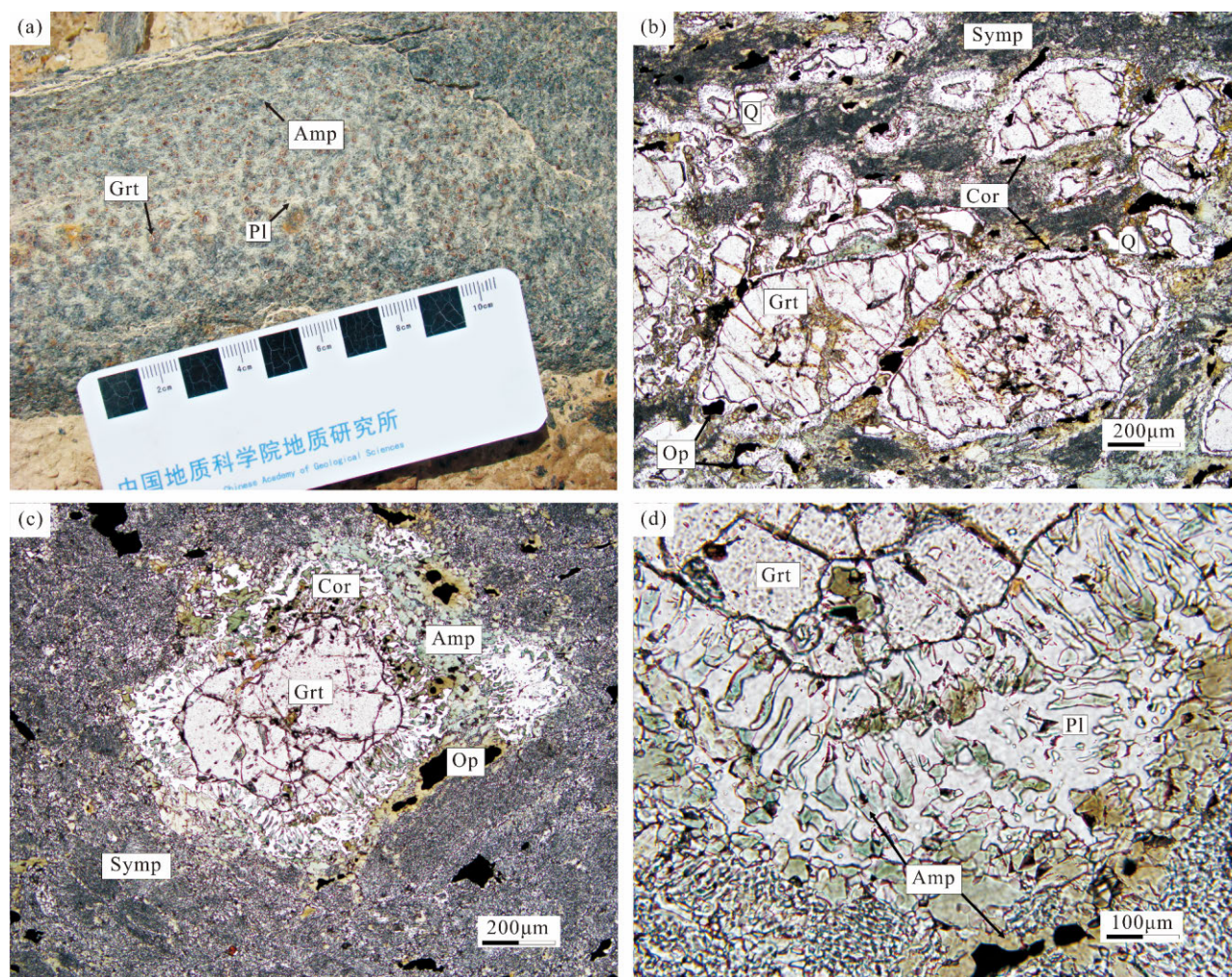


图2 夏日哈木退变榴辉岩野外照片及显微特征

Fig 2 The field photograph and microphotographs of the retrograded eclogite from the Xiarihamu area

(a)一野外露头,可见白眼圈结构;(b)一退变榴辉岩:岩石呈变斑杂状,变斑晶主要为粒度不等的石榴子石,基质由角闪石+斜长石构成的后成合晶、石英以及不透明矿物组成,石榴子石外围具有宽度不等的“白眼圈”;(c)一石榴子石具“白眼圈”结构,被角闪石+斜长石构成的后成合晶包围;(d)一石榴子石具指纹状交生的冠状体(白眼圈),主要由针柱状角闪石和他形斜长石构成,外围被灰绿色角闪石和后成合晶包围;Grt—石榴子石;Amp—角闪石;Pl—斜长石;Q—石英;Op—不透明矿物;Symp—后成合晶;Cor—冠状体

(a)—The field outcrop of the retrograde eclogite with “white eyes” texture; (b)—overview of the retrograde eclogite: porphyroblastic texture, phenocrysts are mainly multi-granularity garnets with “white eye” and matrix composed of hornblende-plagioclase symplectite, quartz and opaque minerals; (c)—coronitic garnets set in grey-green amphiboles + white plagioclases symplectite matrix; (d)—the corona consisting of intimate finger-like intergrowths of grey-green amphiboles and plagioclases along the rim of garnet; Grt—garnet; Amp—amphibole; Pl—plagioclase; Q—quartz; Op—opaque mineral; Symp—symplectite; Cor—corona

土元素和微量元素采用等离子质谱法(ICP-MS)测得。氧化物和微量元素含量大于 10×10^{-6} 的元素分析误差为 5%; 小于 10×10^{-6} 的元素分析误差为 10%(Zeng Lingsen et al., 2011)。

Sm-Nd 同位素组成在中国地质科学院地质研究所同位素实验室测定。Sm-Nd 同位素测定采用同位素稀释法, 仪器为 MAT262 固体同位素质谱仪。Nd 同位素组成测定在 Nu Plasam HR MC-ICP-MS (Nu Instruments) 完成; 标样为 JMC Nd₂O₃, 标样的 $^{143}\text{Nd}/^{144}\text{Nd} = 0.511126 \pm 10(2\sigma)$; Nd 同位素质量分馏采用 $^{146}\text{Nd}/^{144}\text{Nd} = 0.7219$ 校正。详细分析流程见 He Xuexian et al. (2007)。

4 结果

4.1 全岩地球化学

4.1.1 主量元素

夏日哈木榴辉岩的主量元素分析结果见表 1。样品 SiO₂ 含量变化于 46.3%~50.8%, 属于基性岩石。在 Zr/Ti-Nb/Y 图解上, 所有样品位于亚碱性玄武岩范围(图 3a), 在 AFM 图解上, 所有样品都位于拉斑玄武岩范围(图 3b)。MgO 含量为 5.64%~9.13%, FeO^t 含量为 10.14%~15.51%, Al₂O₃ 含量为 13.15%~15.13%, CaO 含量为 9.66%~12.95%。Na₂O、K₂O 含量较低, 分别为 1.68%~2.59%、0.07%~0.42%。Na₂O/K₂O 比值变化较大, 为 1.2~37.7, 总体显示富 Na 的特点。TiO₂ 含量变化较大, 为 0.82%~3.14%, 全部高于岛弧拉斑玄武岩 TiO₂ 含量(0.8%, Wilson, 1989), 其中有两件样品的 TiO₂ 含量高于洋岛拉斑玄武岩 TiO₂ 含

量(2.63%, Wilson, 1989), 具有高钛榴辉岩(>2%)富 Fe、富 P、贫 Mg、贫 Ca、Al₂O₃/TiO₂ 比值偏低的特点(Qiu Jiansheng et al., 2006)。

4.1.2 微量元素

夏日哈木榴辉岩的稀土微量元素分析结果见表 1。根据 Sajona et al. (1993, 1996) 对富 Nb 玄武岩(Nb-Enriched Basalt, NEB) 的定义($\text{Nb} > 7 \times 10^{-6}$), 夏日哈木榴辉岩部分样品(表 1)也具有富 Nb 的特点, 故此本文将夏日哈木榴辉岩划分为富 Nb 型榴辉岩($\text{Nb} > 7 \times 10^{-6}$)和正常榴辉岩($\text{Nb} < 7 \times 10^{-6}$)(图 4)。

富 Nb 型榴辉岩的稀土元素总量较高, ΣREE 为 $118 \times 10^{-6} \sim 148 \times 10^{-6}$, 是 CI 球粒陨石稀土元素总量的 47~60 倍(Sun and McDough, 1989), (La/Sm)_N 比值为 1.31~1.59, (La/Yb)_N 比值为 1.56~3.65, δEu 值为 0.92~0.96; 正常榴辉岩的稀土元素总量较低($\Sigma\text{REE} = 25 \times 10^{-6} \sim 68 \times 10^{-6}$), 是 CI 球粒陨石稀土元素总量的 10~26 倍, (La/Sm)_N 比值为 0.89~1.06, (La/Yb)_N 比值为 0.94~1.09, δEu 值为 0.93~1.09。在稀土元素配分模式图中, 富 Nb 型榴辉岩轻重稀土元素分馏程度稍高, 配分曲线呈轻稀土元素弱富集的右倾型, 表现出 OIB 的亲缘性, 与古生代富 Nb 玄武岩(NEB) 和 Seki 玄武岩稀土元素配分模式类似(图 5a), 而正常榴辉岩轻重稀土元素分馏较弱, 配分曲线呈平坦型, 表现出 E-MORB 的亲缘性, 与 Okinawa 陆内弧后盆地玄武岩的配分模式相似(图 5b)。两者均未表现出 Eu 的异常(图 5a, b)。在微量元素蛛网图上, 富 Nb 型榴辉岩表现出弱的 Nb、Ta、Sr 异常, 与

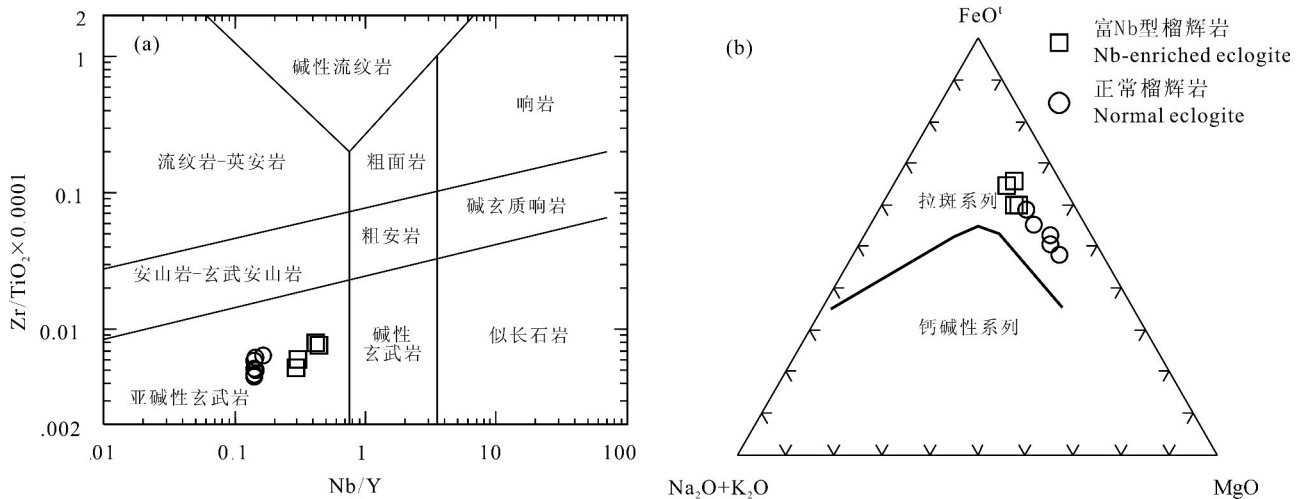


图 3 夏日哈木榴辉岩石类型划分图解(a 据 Pearce, 1996; b 据 Irvine and Baragar, 1971)

Fig. 3 The classification graphs of eclogites from Xiarihamu area (a after Pearce, 1996; b after Irvine and Baragar, 1997)

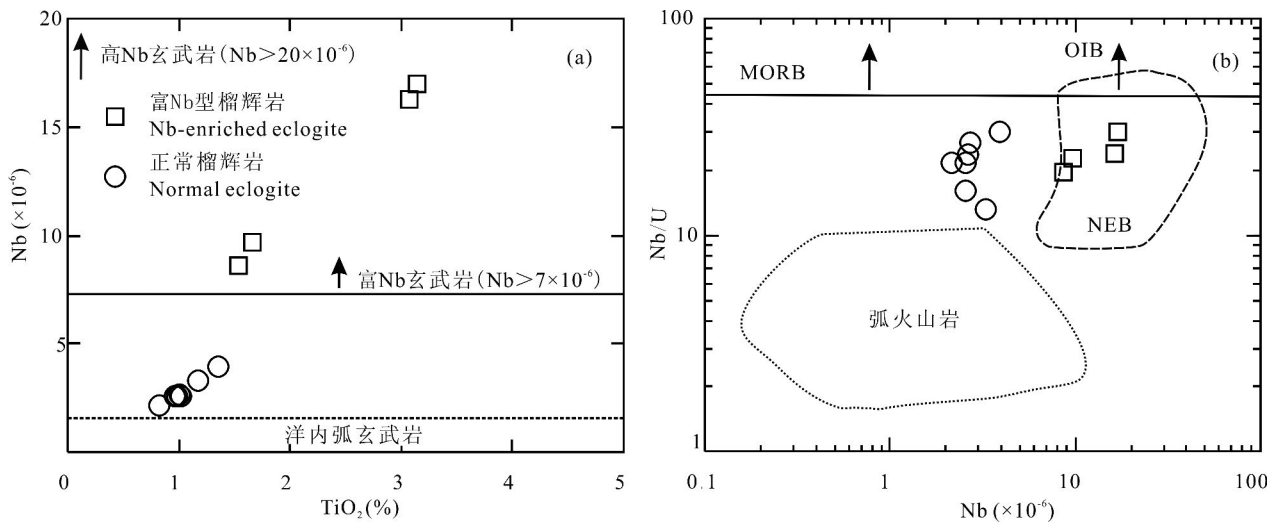
表 1 东昆仑夏日哈木榴辉岩的主量(%)、微量元素($\times 10^{-6}$)分析结果

Table 1 Major (%) and trace element ($\times 10^{-6}$) compositions of the Xiarihamu eclogites from East Kunlun

序号	1	2	3	4	5	6	7	8	9	10	11
样号	K14-4-3. 5	K14-4-3. 6	K14-4-3. 10	K14-4-3. 11	K14-4-3. 1	K14-4-4. 2	K14-4-4. 4	K14-4-4. 7	K14-4-4. 13	K14-4-4. 15	K14-4-4. 20b
类型	富 Nb 型榴辉岩				正常榴辉岩						
SiO ₂	46. 29	48. 77	47. 4	48. 56	49. 68	49. 11	49. 49	50. 78	48. 88	49. 74	48. 71
TiO ₂	1. 66	3. 07	1. 54	3. 14	1	1. 02	0. 82	1. 36	0. 95	1. 17	1. 01
Al ₂ O ₃	14. 02	13. 15	14. 08	13. 9	14. 86	14. 89	14. 06	14. 63	15. 13	13. 86	14. 65
Fe ₂ O ₃	3. 61	1. 85	3. 6	3. 55	4. 11	2. 31	2. 42	2. 87	2. 92	2. 76	2. 47
FeO	9. 97	13. 85	10. 94	12. 16	7. 42	8. 93	7. 96	10. 44	8. 03	9. 54	8. 86
MnO	0. 23	0. 22	0. 28	0. 22	0. 19	0. 19	0. 17	0. 22	0. 18	0. 21	0. 19
MgO	6. 15	5. 84	6. 75	5. 64	8. 24	8. 26	9. 13	6. 8	8. 3	7. 4	8. 44
CaO	12. 95	9. 94	11. 84	9. 72	12. 55	12. 4	12. 8	9. 66	12. 85	11. 8	12. 1
Na ₂ O	2. 59	1. 99	2. 47	2. 27	1. 68	1. 74	1. 72	2. 26	1. 9	2. 13	1. 91
K ₂ O	0. 27	0. 23	0. 26	0. 42	0. 08	0. 13	0. 14	0. 06	0. 07	0. 15	0. 21
P ₂ O ₅	0. 18	0. 34	0. 19	0. 31	0. 08	0. 07	0. 05	0. 1	0. 07	0. 12	0. 07
LOI	2. 28	0	1. 12	0	0. 37	0. 69	1. 41	0. 43	0. 74	1. 76	1. 6
Total	100. 2	99. 25	100. 47	99. 89	100. 26	99. 74	100. 17	99. 61	100. 02	100. 64	100. 22
Mg [#]	45. 6	40. 4	46. 1	39. 8	57. 2	57. 5	61. 8	48. 5	58. 4	52. 6	57. 8
La	9. 79	20. 6	7. 96	14. 9	2. 89	2. 92	2. 21	4. 42	3. 29	3. 95	2. 95
Ce	22. 7	48. 4	18. 3	35. 5	7. 54	8	5. 78	11. 6	8. 16	10	7. 62
Pr	3. 14	6. 63	2. 51	5. 15	1. 18	1. 24	0. 89	1. 67	1. 22	1. 51	1. 19
Nd	15. 8	32. 4	12. 8	25. 6	6. 51	6. 61	4. 72	8. 39	6. 94	8. 06	6. 47
Sm	4. 36	8. 13	3. 83	7. 01	2. 04	2. 04	1. 46	2. 62	2. 08	2. 48	1. 95
Eu	1. 51	2. 58	1. 38	2. 27	0. 79	0. 82	0. 53	1. 08	0. 88	0. 93	0. 77
Gd	5. 31	8. 9	5. 26	8. 12	2. 72	2. 8	2. 08	3. 78	2. 94	3. 41	2. 68
Tb	0. 85	1. 29	0. 88	1. 33	0. 5	0. 52	0. 37	0. 66	0. 53	0. 58	0. 48
Dy	5. 88	8. 24	5. 85	8. 25	3. 45	3. 6	2. 77	4. 82	3. 65	4. 23	3. 24
Ho	1. 15	1. 47	1. 08	1. 38	0. 67	0. 65	0. 54	0. 85	0. 65	0. 8	0. 67
Er	3. 29	4. 3	3. 41	3. 88	2. 06	2. 07	1. 6	2. 47	2. 07	2. 45	1. 93
Tm	0. 55	0. 61	0. 52	0. 56	0. 31	0. 3	0. 24	0. 41	0. 3	0. 39	0. 3
Yb	3. 6	3. 8	3. 44	3. 42	2. 02	2. 02	1. 58	2. 56	1. 99	2. 48	1. 97
Lu	0. 53	0. 54	0. 54	0. 54	0. 32	0. 32	0. 25	0. 43	0. 29	0. 38	0. 29
ΣREE	78. 46	147. 89	67. 76	117. 91	33	33. 91	25. 02	45. 76	34. 99	41. 65	32. 51
La _N /Sm _N	1. 41	1. 59	1. 31	1. 34	0. 89	0. 9	0. 95	1. 06	0. 99	1	0. 95
La _N /Yb _N	1. 83	3. 65	1. 56	2. 94	0. 96	0. 97	0. 94	1. 16	1. 11	1. 07	1. 01
δEu	0. 96	0. 93	0. 94	0. 92	1. 03	1. 05	0. 93	1. 05	1. 09	0. 98	1. 03
Y	31. 4	39. 5	30	38. 6	18	17. 8	14. 8	23. 7	17. 4	23. 1	17. 7
Rb	6. 83	5. 95	7. 21	10. 3	2. 18	4. 8	3. 3	2. 03	1. 56	3. 41	9. 44
Sr	200	240	158	225	155	148	105	27. 6	135	170	159
Ba	127	73. 8	78. 9	150	10. 2	26. 8	33. 1	8. 56	10. 9	63. 1	43. 4
Nb	9. 67	16. 3	8. 63	17	2. 57	2. 61	2. 14	3. 92	2. 57	3. 28	2. 69
Ta	0. 64	1. 14	0. 6	1. 08	0. 19	0. 18	0. 15	0. 26	0. 18	0. 23	0. 17
Zr	104	235	87. 7	240	53. 6	52. 2	41. 5	86. 6	61. 5	60. 8	55. 2
Hf	2. 99	6. 68	2. 66	6. 62	1. 65	1. 61	1. 15	2. 61	1. 88	1. 74	1. 55
Th	1. 22	2. 59	0. 93	1. 89	0. 33	0. 32	0. 29	0. 54	0. 4	0. 45	0. 33
U	0. 43	0. 69	0. 44	0. 57	0. 12	0. 11	0. 1	0. 13	0. 16	0. 25	0. 1
V	380	366	335	385	272	280	251	342	287	322	284
Cr	125	95. 7	111	83	211	215	303	102	302	125	228
Co	46. 1	52. 1	51. 6	52. 1	50. 8	47. 7	45. 8	51. 9	43. 7	50	53
Ni	64	69. 6	67. 4	69. 2	98. 4	100	101	59. 5	98. 7	72. 3	107
Sc	44. 3	34. 2	41. 3	34. 2	42. 9	40	39. 8	46. 5	43	44. 9	41. 9

注: Mg[#] = Mg / (Mg + Fe²⁺) × 100; 标准化数据根据 Sun and McDougall (1989); δEu = Eu_N × 2 / (Sm_N + Gd_N)。

古生代 NEB 类似(图 5c); 正常榴辉岩表现出富 U 岩相比, Nb、Ta 亏损程度较低, 而 Sr 变化较大(图 5d)。贫 Nb、Ta 的特点, 与 Okinawa 陆内弧后盆地玄武

图4 夏日哈木榴辉岩 Nb-TiO₂ (a) 和 Nb/U-Nb (b) 图解Fig. 4 TiO₂ vs Nb (a) and Nb vs Nb/U (b) diagrams for Xiarihamu eclogites

(a)—高 Nb 玄武岩和富 Nb 玄武岩数据来自 Sajona et al., 1993, 1996, 洋内弧玄武岩数据来自 Martin et al., 2005;

(b)—弧火山岩和富 Nb 玄武岩数据来自 Kepezhinaskas et al., 1996

(a)—High-Nb basalt and Nb-enriched basalt data from Sajona et al., 1993, 1996, intraoceanic arc basalt after Martin et al., 2005;

(b)—arc volcanic rocks data and Nb-enriched basalt (NEB) data from Kepezhinaskas et al., 1996

4.2 Sm-Nd 同位素

本文对夏日哈木 11 件榴辉岩样品进行了 Sm、Nd 同位素组成分析(表 2)。富 Nb 型榴辉岩的 Sm、Nd 含量分别为 $3.87 \times 10^{-6} \sim 7.68 \times 10^{-6}$ 、 $11.9 \times 10^{-6} \sim 36.1 \times 10^{-6}$, $^{147}\text{Sm}/^{144}\text{Nd}$ 比值为 $0.1244 \sim 0.2555$, $^{147}\text{Nd}/^{144}\text{Nd}$ 比值为 $0.512638 \sim 0.512871$; 正常榴辉岩的 Sm、Nd 含量分别为 $1.47 \times 10^{-6} \sim 2.79 \times 10^{-6}$ 、 $4.83 \times 10^{-6} \sim 9.6 \times 10^{-6}$,

$^{147}\text{Sm}/^{144}\text{Nd}$ 比值为 $0.1561 \sim 0.1906$, $^{147}\text{Nd}/^{144}\text{Nd}$ 比值为 $0.512751 \sim 0.512904$ 。富 Nb 型榴辉岩具有较高的 Sm、Nd 含量, 变化较大, 计算的 $\epsilon_{\text{Nd}}(0)$ 值为 $0 \sim 4.55$, $\epsilon_{\text{Nd}}(t)$ 值为 $-0.88 \sim 7.26$ ($t = 571\text{Ma}$, Qi Shengsheng et al., 2014); 正常榴辉岩的 Sm、Nd 含量较低, 变化较小, 计算的 $\epsilon_{\text{Nd}}(0)$ 值为 $2.20 \sim 5.19$, $\epsilon_{\text{Nd}}(t)$ 值为 $3.17 \sim 6.22$ ($t = 571\text{Ma}$, Qi Shengsheng et al., 2014)。

表 2 东昆仑夏日哈木榴辉岩 Sm-Nd 同位素组成

Table 2 The Sm-Nd isotopic compositions of the Xiarihamu eclogites from East Kunlun

样号	Sm($\times 10^{-6}$)	Nd($\times 10^{-6}$)	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	2σ	$f_{\text{Sm}/\text{Nd}}$	$\epsilon_{\text{Nd}}(0)$	$\epsilon_{\text{Nd}}(t)$
K14-4-3.1	2.03	7.85	0.1561	0.512805	0.000018	-0.21	3.26	6.22
K14-4-3.5	4.34	11.90	0.2206	0.512682	0.000012	0.12	0.86	-0.88
K14-4-3.6	7.43	36.10	0.1244	0.512740	0.000007	-0.37	1.99	7.26
K14-4-3.10	3.87	18.30	0.1277	0.512638	0.000005	-0.35	0.00	5.03
K14-4-3.11	7.68	18.20	0.2555	0.512871	0.000007	0.30	4.55	0.26
K14-4-4.2	2.04	6.98	0.1765	0.512766	0.000035	-0.10	2.50	3.97
K14-4-4.4	1.47	4.83	0.1835	0.512751	0.000013	-0.07	2.20	3.17
K14-4-4.7	2.79	9.60	0.1760	0.512790	0.000009	-0.11	2.97	4.47
K14-4-4.13	2.05	6.50	0.1906	0.512904	0.000008	-0.03	5.19	5.63
K14-4-4.15	2.60	8.37	0.1875	0.512796	0.000007	-0.05	3.08	3.75
K14-4-4.20b	2.08	6.65	0.1891	0.512872	0.000009	-0.04	4.56	5.12

注: $f_{\text{Sm}/\text{Nd}} = (^{147}\text{Sm}/^{144}\text{Nd})_{\text{sample}} / 0.1967 - 1$; $\epsilon_{\text{Nd}}(0) = \left[(^{143}\text{Nd}/^{144}\text{Nd})_{\text{sample}} / 0.512638 - 1 \right] \times 10^4$; $\epsilon_{\text{Nd}}(t) = \epsilon_{\text{Nd}}(0) - Q \times f_{\text{Sm}/\text{Nd}} \times t$ ($Q = 25.1\text{Ga}^{-1}$); $t = 571\text{Ma}$ (Qi Shengsheng et al., 2014)。

5 讨论

5.1 榴辉岩原岩性质

榴辉岩在复杂的进变质和退变质过程中, 一些

化学元素尤其是碱性元素比较活跃, 容易随着流体迁移 (Ling Wenli et al., 2000; Jahn et al., 2003), 但是岩石主量元素在榴辉岩变质过程中的变化是很小的, 例如 TiO_2 、 P_2O_5 、 Al_2O_3 在蚀变中基本保持稳

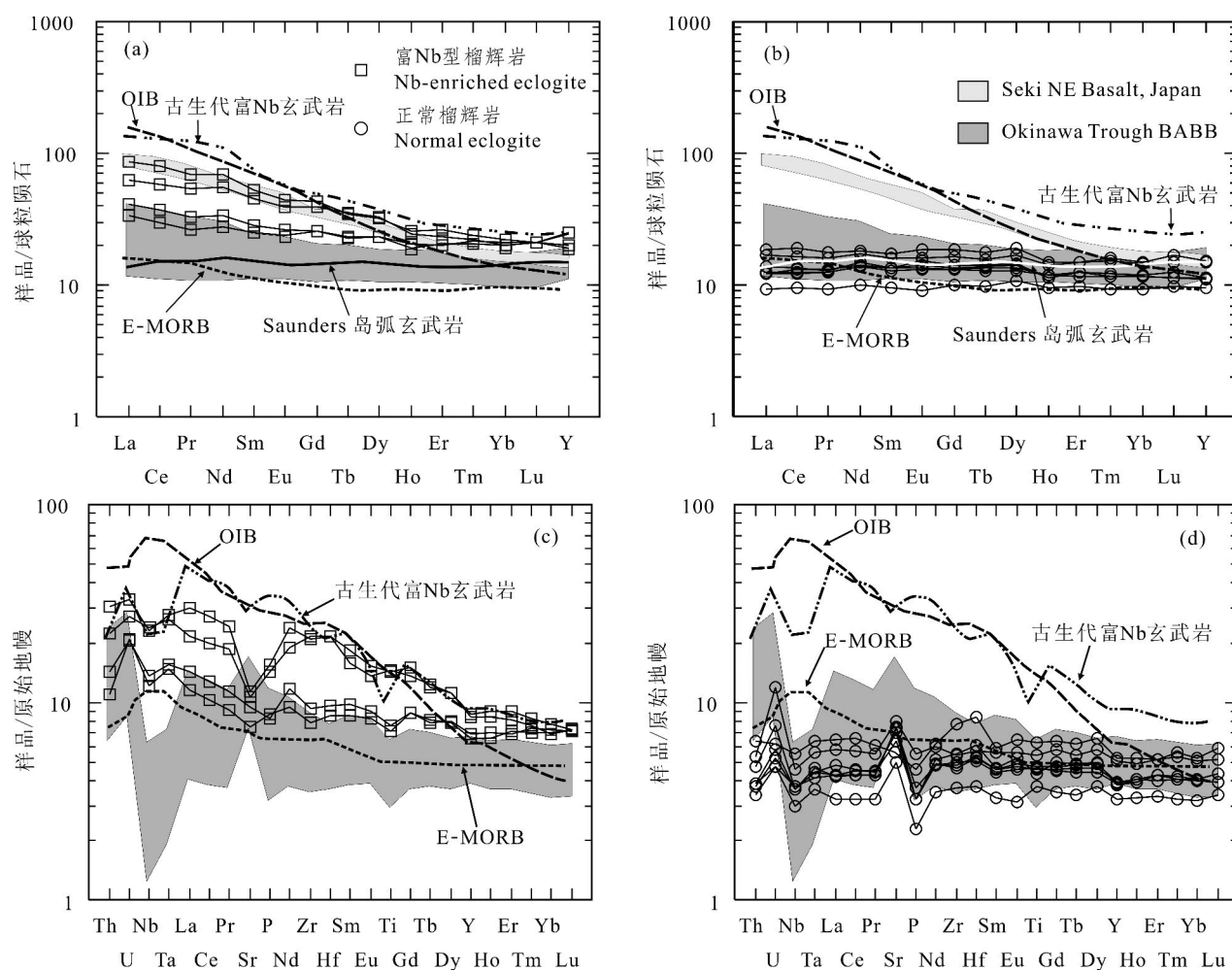


图5 夏日哈木榴辉岩稀土(a, b)、微量元素(c, d)配分模式图

Fig. 5 Chondrite-normalized rare earth elements distributions (a, b) and primitive mantle-normalized spider diagram for the Xiarihamu eclogites (c, d)

古生代富Nb玄武岩据 Zhang Haixiang et al., 2005; Saunders 岛弧玄武岩据 Pearce et al., 1995; Okinawa 弧后盆地玄武岩(BABB)

据 Shinjo et al., 1999; Seki 富Nb玄武岩据 Shuto et al., 2006; OIB, E-MORB, N-MORB 据 Sun and McDonough, 1989

Paleozoic NEB from Zhang Haixiang et al., 2005; Saunders island-arc basalt from Pearce et al., 1995; Okinawa Trough BABB

from Shinjo et al., 1999; Seki basalt from Shuto et al., 2006; OIB, E-MORB, N-MORB all from Sun and McDonough, 1989

定,通常能够保留原岩的性质和类型(Bernard-Griffiths et al., 1985)。此外,稀土元素在榴辉岩相变质过程中也可以看做是稳定的(Bernard-Griffiths et al., 1985; Shatsky et al., 1990)。北秦岭榴辉岩的研究表明,在退变质过程中虽然有流体活动,如石英脉切割榴辉岩,榴辉岩的 HSFE、REE 也没有明显的迁移,只对 Rb 和 Ba 有一定影响(Wang Hao et al., 2013)。因此,可以利用这些稳定元素(主量元素、REE、HSFE)讨论榴辉岩的原岩性质(Peltonen et al., 1996)。夏日哈木榴辉岩的 $Mg^\#$ 值范围在 40~62,显然其原岩不是原生岩浆直接结晶的产物(68~75, Frey et al., 1978),尤其是富 Nb 型榴辉岩,经过 CIPW 计算含有一定量的石英标准矿物,这与大

别山毛屋和碧溪岭榴辉岩类似,表明原岩可能是分异程度较高的岩体(Jahn et al., 2003)。西天山榴辉岩是大洋地壳俯冲变质的产物(Gao Jun and Klemd, 2003),柴北缘北带、温泉、碧溪岭榴辉岩是就位于大陆地壳的基性侵入体或火山岩(Chavagnac and Jahn, 1996; Song Shuguang et al., 2003; Meng Fancong et al., 2013)。在主量元素相关图上(图 6), Al_2O_3 、 FeO^* 和 TiO_2 显示 MORB 的亲性,而 Na_2O 具有 MORB 和弧后盆地玄武岩的亲性,同时与温泉、柴北缘北带榴辉岩主量元素特征相似。

夏日哈木榴辉岩的稀土元素特征具有 E-MORB、OIB 和 NEB 的亲缘性(图 5a, b),类似于都

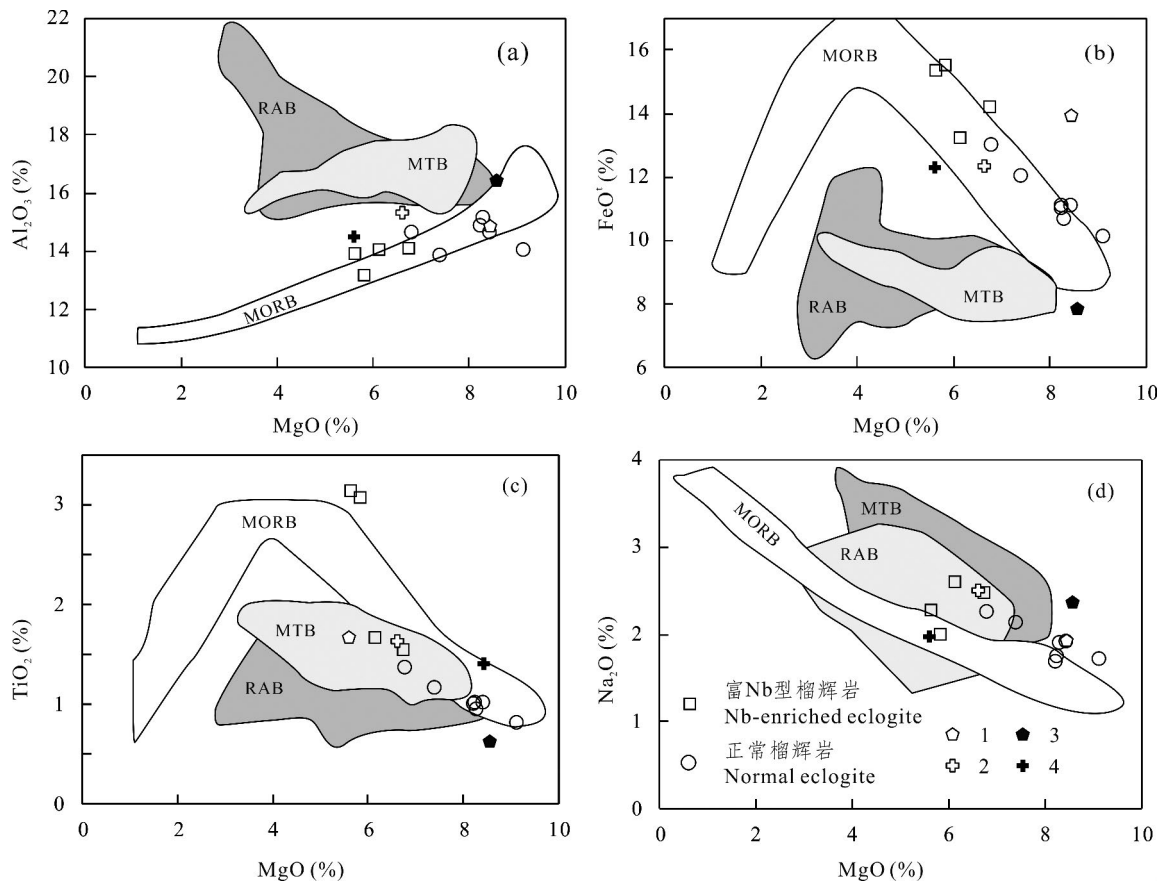


图6 夏日哈木榴辉岩主量元素相关图

Fig. 6 The covariant relation of major elements of the Xiarhamu eclogites

1—柴北缘北带榴辉岩化学组成平均值(Song Shuguang et al., 2003); 2—温泉榴辉岩化学组成平均值(Meng Fancong et al., 2015); 3—碧溪岭榴辉岩化学组成平均值(Chavagnac and Jahn, 1996); 4—西天山榴辉岩化学组成平均值(Gao Jun and Klemd, 2003); MORB—大洋地壳组成; RAB—陆内弧后盆地玄武岩; MTB—洋内弧后盆地玄武岩; MORB、RAB(Ryukyu Arc Basalt)、MTB(Mariana Trough Basalt)范围来自 Hawkins(1995)和 Shinjo et al. (1999)

1—Eclogite from north belt of North Qaidam (Song Shuang et al., 2003); 2—eclogite from Wenquan (Meng Fancong et al., 2015); 3—eclogite from Bixiling (Chavagnac and Jahn, 1996); 4—eclogite from West Tianshan (Gao Jun and Klemd, 2003); fields shown are MORB, Ryukyu Arc Basalt (RAB) and Mariana Trough Basalt (MTB) constructed by Hawkins (1995) and Shinjo et al. (1999)

兰榴辉岩稀土元素特征(Song Shuguang et al., 2003; Yu Shengyao et al., 2013),但未出现温泉细粒榴辉岩轻稀土元素亏损的特征(Meng Fancong et al., 2015)。富Nb型榴辉岩稀土元素配分曲线与古生代富Nb玄武岩(NEB)和Seki玄武岩的配分曲线是一致的(图5a),说明可能是受到了俯冲板片熔体的影响。Sato et al. (2007)认为在大洋地壳俯冲过程中,诱发对流的软流圈与岩石圈相互作用,在弧后盆地附近可以形成具有富Nb、富Ti等地球化学特征的玄武岩,如Seki玄武岩(Shuto et al., 2006),这与富Nb型榴辉岩地球化学特征相似。正常榴辉岩的稀土元素配分曲线(图5a)与Okinawa陆内弧后盆地玄武岩(BABB, Shinjo et al., 1999)的稀土元素配分曲线是一致的,说明夏日哈木榴辉岩

的原岩可能是来自于陆内弧后盆地玄武岩,源区可能存在俯冲板片熔体的贡献,这种现象可以与中国东海和琉球群岛玄武岩对比(Shinjo et al., 1999)。在Nb/Th-La/Nb图解上,所有样品均具有弧火山岩或地壳混染的趋势(图7a)(Zhang Aimei et al., 2012);在Ce/Nb-Th/Nb图解上,富Nb型榴辉岩位于OIB范围,正常榴辉岩位于日本海和板内弧后盆地之间区域(图7b)(Shinjo et al., 1999);在Th/Yb-Nb/Yb图解上,所有样品都位于MORB和OIB阵列中,并存在向地壳过渡的趋势,说明可能存在同化混染下地壳(图7c)(Pearce et al., 1995);在Nb/Zr-Th/Zr图解上,样品位于勘察加半岛Valovayam/Tymlat埃达克岩-富Nb玄武岩范围内,具有熔体富集的趋势,可能是源区受到俯冲板片

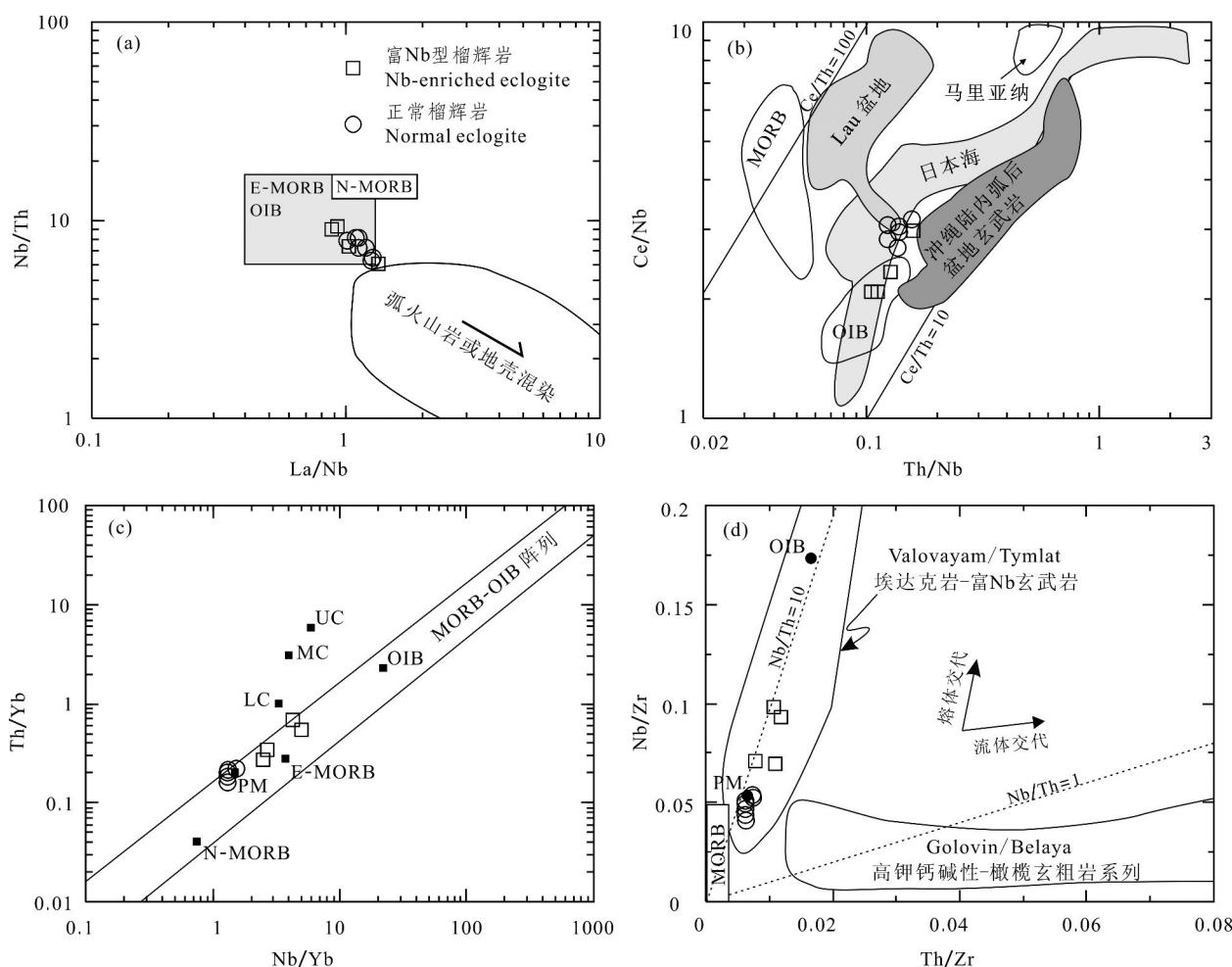


图7 夏日哈木榴辉岩 Nb/Th-La/Nb (a)、Ce/Nb-Th/Nb (b)、Th/Yb-Nb/Yb (c)和 Nb/Zr-Th/Zr (d)图解
Fig. 7 Nb/Th vs La/Nb (a), Ce/Nb vs Th/Nb (b), Th/Yb vs Nb/Yb (c) and Nb/Zr vs Th/Zr (d) diagrams of the Xiarihamu eclogites

(a)—据 Zhang Aimei et al., 2012; OIB, N-MORB, E-MORB 数据来自 Sun and McDougall, 1989; (b)—据 Sandeman et al., 2006; Taylor and Martinez, 2003; (c)—据 Pearce et al., 1995; UC, MC, LC—上、中、下地壳数据源自 Rudnick and Fountain, 1995; PM, N-MORB, OIB, E-MORB 源自 Sun and McDougall, 1989; (d) 据 Kepezhinskis et al., 1997

(a)—after Zhang Aimei et al., 2012; OIB, N-MORB, E-MORB from Sun and McDougall, 1989; (b)—after Sandeman et al., 2006; Taylor and Martinez, 2003; (c)—after Pearce et al., 1995; data of the UC, MC, LC—Upper, Middle, Low Crust from Rudnick and Fountain, 1995; data of PM, N-MORB, OIB, E-MORB from Sun and McDougall, 1989; (d) after Kepezhinskis et al., 1997

熔体的改造(图 7d)(Kepezhinskis et al., 1997)。

Sm-Nd 同位素是识别 UHP 和 HP 变质岩原岩性质非常重要的方法(Jahn, 1999)。Jahn(1999)系统收集了阿尔卑斯期、海西期、加里东期和大别-苏鲁等 HP-UHP 变质带内榴辉岩、石榴橄榄岩的 Sm-Nd 同位素数据,认为来自阿尔卑斯造山带(除了 Dora-Maira 含柯石英变沉积岩)和海西造山带的 UHP 变质岩具有大洋地壳的亲缘性,而加里东造山带和大别苏鲁造山带的 UHP 变质岩具有大陆地壳的亲缘性。在 $\epsilon_{\text{Nd}}(0) - {}^{147}\text{Sm}/{}^{144}\text{Nd}$ 和 $\epsilon_{\text{Nd}}(t) - {}^{147}\text{Sm}/{}^{144}\text{Nd}$ 图解上均位于加里东 HP-UHP 变质带和苏鲁-大别 HP-UHP 变质带 Sm-Nd 同位素组成

范围内,与柴北缘北带陆壳型榴辉岩同位素组成类似(图 8),因此,夏日哈木榴辉岩的原岩可能是具有陆壳性质的玄武质岩石。

5.2 年代学

前人对夏日哈木榴辉岩的峰期变质年龄和榴辉岩退变质年龄做了较多的工作。Qi Shengsheng et al. (2014) 获得夏日哈木榴辉岩锆石 LA-ICP-MS U-Pb 年龄为 $411 \pm 2\text{Ma}$,其中三颗锆石给出 $499 \sim 571\text{Ma}$ 的年龄,可能代表原岩时代。Wang Guan (2014)初步确定透镜状石榴子石斜长角闪岩原岩为具有 E-MORB 特征的洋壳,榴辉岩相变质年龄为 $438 \pm 4\text{Ma}$ 。Meng Fancong et al. (2013)获得温泉

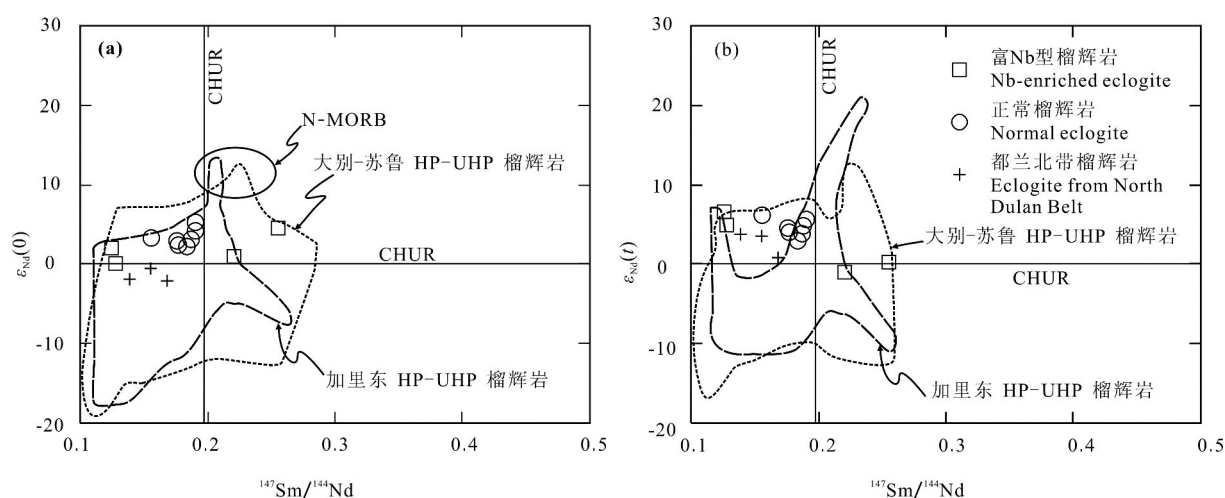


图 8 夏日哈木榴辉岩 $\epsilon_{\text{Nd}}(0)$ - $^{147}\text{Sm}/^{144}\text{Nd}$ (a) 和 $\epsilon_{\text{Nd}}(t)$ - $^{147}\text{Sm}/^{144}\text{Nd}$ (b) 图解

Fig. 8 $\epsilon_{\text{Nd}}(0)$ vs $^{147}\text{Sm}/^{144}\text{Nd}$ (a) and $\epsilon_{\text{Nd}}(t)$ vs $^{147}\text{Sm}/^{144}\text{Nd}$ (b) diagrams of the Xiarihamu eclogites

大别-苏鲁 HP-UHP 榴辉岩、加里东 HP-UHP 榴辉岩同位素数据来自 Jahn, 1999 及其参考文献; 都兰北带榴辉岩数据来自 Song Shuguang et al., 2003; N-MORB 数据来自 Spencer et al. 1995; CHUR 数据来自 Jacobsen and Wasserburg, 1980, ($^{147}\text{Sm}/^{144}\text{Nd}$)_{CHUR} = 0.1967, ($^{143}\text{Nd}/^{144}\text{Nd}$)_{CHUR} = 0.512638; $t = 571\text{Ma}$ (Qi Shengsheng et al., 2014)

Sm and Nd isotopic data of the Dabie-Sulu HP-UHP belt and the Caledonian HP-UHP belt from Jahn, 1999 and its references; Sm and Nd isotopic data of North Dulan Belt from Song Shuguang et al., 2003; N-MORB data from Spencer et al. 1995; CHUR from Jacobsen and Wasserburg, 1980, ($^{147}\text{Sm}/^{144}\text{Nd}$)_{CHUR} = 0.1967, ($^{143}\text{Nd}/^{144}\text{Nd}$)_{CHUR} = 0.512638, $t = 571\text{Ma}$ (Qi Shengsheng et al., 2014)

榴辉岩的 $^{206}\text{Pb}/^{238}\text{U}$ 加权年龄为 $428 \pm 2\text{Ma}$, 然而 Jia Lihui et al. (2014) 获得温泉榴辉岩中高压石英脉锆石 U-Pb 年龄为 $451 \pm 2\text{Ma}$ 。Qi Xiaopeng et al. (2016) 在郎木日上游发现的榴辉岩, 其变质年龄为 $431.9 \pm 2.3\text{Ma}$ 。Du Wei et al. (2017) 获得大格勒榴辉岩变质锆石 SHRIMP U-Pb 年龄为 $433 \pm 5\text{Ma}$ 。根据夏日哈木含矿基性—超基性岩体未发生变质的事实, 可以认为夏日哈木榴辉岩的变质时代应早于含矿基性—超基性岩的年龄 ($393.5 \sim 439.1\text{Ma}$, Li Shijin et al., 2012; Li Chusi et al., 2015; Jiang Changyi et al., 2015; Song Xieyan et al., 2016; Peng Bo et al., 2016; Feng Chenyou et al., 2016; Liu Jiannan et al., 2017; Zhang Zhaowei et al., 2017)。由此可见, 夏日哈木榴辉岩的峰期变质可能发生在晚奥陶世—志留纪。

目前, 东昆仑西段夏日哈木榴辉岩原岩形成时代研究还是空白。根据区域地质资料, 其原岩形成时代存在两种可能性: 一种可能是形成于新元古代早期, 与温泉榴辉岩类似 (锆石 U-Pb 年龄 $934 \pm 11\text{Ma}$, Meng Fancong et al., 2013); 另一种可能是形成于新元古代晚期—早古生代。新元古代早期的构造热事件是东昆仑地区前寒武纪最强烈的一次构造聚合事件 ($\sim 1000\text{Ma}$, Wang Guocan et al., 2007), 与全球尺度的 Rodinia 超大陆的形成时间是

吻合的 (Condie, 2001), 此次构造热事件使东昆仑地区基底岩系发生强烈的深层次韧性剪切流动构造和广泛的深熔作用 (Wang Guocan et al., 2004; Meng Fancong et al., 2013; Chen Youxin et al., 2015; He Dengfeng et al., 2016), 温泉榴辉岩的原岩是在这样的构造体制下形成的 (Meng Fancong et al., 2013, 2015)。然而, 夏日哈木榴辉岩原岩具有弧后盆地玄武岩的地球化学特征, 说明其与温泉榴辉岩的原岩是在不同的构造体制下形成的。前人对东昆仑地区新元古代—泥盆纪变质岩和岩浆岩做了大量的年代学研究, 由此归纳出东昆仑地区新元古代—早古生代构造演化为: ($>$) $1000 \sim 800\text{Ma}$, 碰撞阶段, 响应 Rodinia 超大陆汇聚事件 (Tan Shengxiang et al., 2004; Meng Fancong et al., 2013; Ma Wen et al., 2013; Chen Youxin et al., 2015; He Dengfeng et al., 2016); $800 \sim ?\text{Ma}$, Rodinia 超大陆裂解, 原特提斯洋形成演化阶段 (Ren Junhu et al., 2010, 2011; Xu Xin et al., 2016); $>550 \sim 460\text{Ma}$, 沟-弧-盆体系形成及演化阶段 (Yang Jingsui et al., 1996; Li Huankun et al., 2006; Ma Yongshou et al., 2010; Zhang Yafeng et al., 2010; Li Huaikun et al., 2006; Feng Jianyun et al., 2010; Chen Youxin et al., 2013; Qi Xiaopeng et al., 2016; Zheng Yong et al., 2016); $460 \sim 430\text{Ma}$,

弧陆碰撞阶段(Meng Fancong et al., 2013; Liu Bin et al., 2013; Wang Guan, 2014; Xiong Fuhao et al., 2016; Qi Xiaopeng et al., 2016; Du Wei et al., 2017)。目前,关于东昆仑弧后盆地形成的起始时间还不确定,而且东昆仑造山带经历过多旋回洋陆转化以及复杂的多弧岛盆系演化(Yin Hongfu et al., 1997; Yin Fuguang et al., 2003),所以夏日哈木榴辉岩的原岩时代还需要进一步研究和确定。尽管原特提斯洋弧后盆地演化时限还不明确,但大体可以确定,东昆仑早古生代高压变质带的形成可能与弧后盆地的闭合有关。

5.3 地质意义

东昆仑造山带是一个具多旋回复杂演化历史的造山带,经历过多旋回洋陆转化以及复杂的陆内演化(Yin Hongfu et al., 1997),主要体现在早古生代和晚古生代—早中生代两期不同的构造演化过程中(Jiang Chunfa et al., 1992; Yang Jingsui et al., 1996; Lu Songnian et al., 2002; Xu Zhiqin et al., 2006; Zhang Yafeng et al., 2010),其中前人对早古生代构造演化的研究,尤其对洋壳俯冲和碰撞造山过程机制的研究相对较弱。昆北地体早前寒武世结晶基底——金水口群由一套麻粒岩相—高角闪岩相的片麻岩、混合岩、斜长角闪岩、大理岩以及低角闪岩相石英岩、片麻岩等组成,被称为昆北型刚性基底,时代主要为古元古代,可能下沿至太古宙(Wang Yunshan et al., 1987; Jiang Chunfa et al., 1992)。目前发现的夏日哈木榴辉岩、温泉榴辉岩、浪木日榴辉岩、大格勒榴辉岩均位于金水口群中,说明昆北地体内部发生了强烈的陆(弧)—陆碰撞事件,同时也表明东昆仑地区存在一条早古生代高压(超高压?)变质带。

Uyeda 等学者认为大洋俯冲到一定程度,会使弧后软流圈地幔发生强烈的对流,造成弧后应力场由挤压型逐渐过渡为拉张型,从而形成陆内弧后盆地(Uyeda and Kanamori, 1979; Uyeda, 1982, 2002)。一般认为,弧后盆地的演化过程受控于大洋地壳的俯冲作用,消亡过程可能与大洋闭合—陆陆碰撞有关(Talwani and Pitman, 1977; Taylor, 2003; Sun Weidong, 2016)。Shuto et al. (2006)和 Sato et al. (2007)研究发现,在西太平洋陆内弧后盆地可以发育类似 Okinawa 和 Seki 的玄武岩。夏日哈木榴辉岩的原岩具有陆内弧后盆地玄武岩性质,说明在原特提斯洋向北俯冲消减过程中,东昆仑地区存在沟—弧—盆体系。多数学者认为牦牛山组磨拉石建造

是早古生代造山作用结束的标志(Pan Yusheng et al., 1996; Xu Zhiqing et al., 2007; Li Rongshe et al., 2007; Zhang Xueting et al., 2007),其下部层序流纹岩的年龄为 400~423Ma(Lu Lu et al., 2010; Zhang Yaoling et al., 2010),由此可见东昆仑早古生代的构造演化可能持续到了早泥盆世。

6 结论

(1)东昆仑夏日哈木榴辉岩呈透镜状分布于古元古代金水口群中, SiO_2 含量 47.4%~50.78%, $\text{Mg}^\#$ 值为 40~62,表明其原岩为分异程度较高的基性岩,属于拉斑玄武岩系列。根据 Nb 含量可以划分为富 Nb 型榴辉岩($\text{Nb}>7\times 10^{-6}$)和正常榴辉岩($\text{Nb}<7\times 10^{-6}$)。

(2)富 Nb 型榴辉岩与古生代富 Nb 玄武岩稀土元素配分模式类似,可能存在俯冲板片熔体的交代作用;正常榴辉岩与 Okinawa Trough 陆内弧后盆地玄武岩类似;两者 Sm、Nd 同位素组成均位于加里东 HP-UHP 变质带和大别苏鲁 HP-UHP 变质带同位素组成范围内。

(3)夏日哈木榴辉岩原岩具有陆内弧后盆地玄武岩的特点,受到一定程度俯冲板片熔体的改造,在就位过程中可能遭受下地壳的同化混染,在晚奥陶世—志留纪时期,发生了弧后盆地的俯冲变质作用。

(4)东昆仑昆北地体存在一条早古生代高压变质带,可能与原特提斯洋弧后盆地的闭合有关。

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The Protoliths of the Xiarihamu Eclogites from the Western Part of East Kunlun and Continent/Arc-Continent Collision

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Abstract

Xiarihamu eclogites occur as lenses in the Proterozoic Jinshuikou Group at the western part of East Kunlun. The eclogites have SiO_2 content varying from 47.4% to 50.78% and $\text{Mg}^\#$ values from 40 to 62, implying that protoliths of the eclogites were highly fractionated mafic rocks and suggesting that they belong to tholeiite series. This study divides the Xiarihamu eclogites into two groups in the light of Nb contents, Nb-enriched eclogites ($\text{Nb} > 7 \times 10^{-6}$) and normal eclogites ($\text{Nb} < 7 \times 10^{-6}$). The Nb-enriched eclogites are characterized by low SiO_2 , MgO, Al_2O_3 and high FeO^t , TiO_2 . The chondrite-normalized REE patterns of the Nb-enriched eclogites display OIB patterns consistent with Paleozoic Nb-enriched basalts (NEB) and Seki island arc basalts. In addition, the primitive mantle-normalized trace elements spider diagrams have weakly negative anomalies of Nb, Ta, Ti, consistent with Paleozoic Nb-enriched basalts (NEB). The normal eclogites ($\text{Nb} < 7 \times 10^{-6}$) are characterized by low FeO^t , TiO_2 and high SiO_2 , MgO, Al_2O_3 . The REE patterns of normal eclogites display E-MORB patterns; the spider diagrams show weakly negative anomalies of Nb, Ta, P and positive anomalies of U, Sr, consistent with continental back arc basin basalt (BABB) like Okinawa BABB. $\epsilon_{\text{Nd}}(0)$ and $\epsilon_{\text{Nd}}(t=571\text{Ma})$ isotopic contents of the Nb-enriched eclogites are $0 \sim 4.55$ and $-0.88 \sim 7.26$ and those of the normal eclogites are $2.2 \sim 5.19$, $3.17 \sim 6.22$, both plotted into the field of HP-UHP belts of Caledonian orogeny and Sulu-Dabie orogeny. Trace element diagrams also show that the samples have been assimilated and contaminated. Therefore, the protoliths of the Xiarihamu eclogites could be interpreted to be derived from intracontinental BABB with the origin reconstructed by subduction-related crust and assimilation-contamination during magmatism. According to the published age data, it can be concluded that high-pressure metamorphic event in the North Kunlun terrane occurred in the Late Ordovician-Silurian and the Xiarihamu eclogite is probably the result of metamorphism owing to an arc-continent collision during closure of the back-arc basin.

Key words: continent/arc-continent collision; eclogite; Nb-enriched basalt; Xiarihamu; East Kunlun